DNA, RNA, Gene Expression, and Techniques

The M-CATS at UK

DNA and Biotechnology

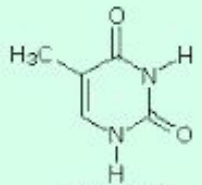
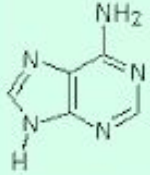
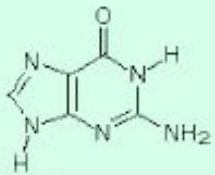
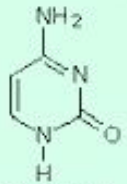
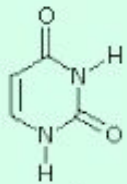
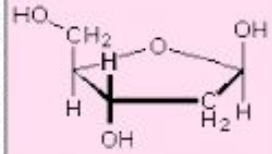
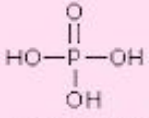
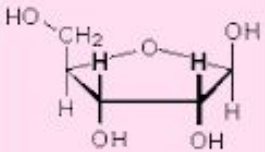
Sugar-Phosphate backbone

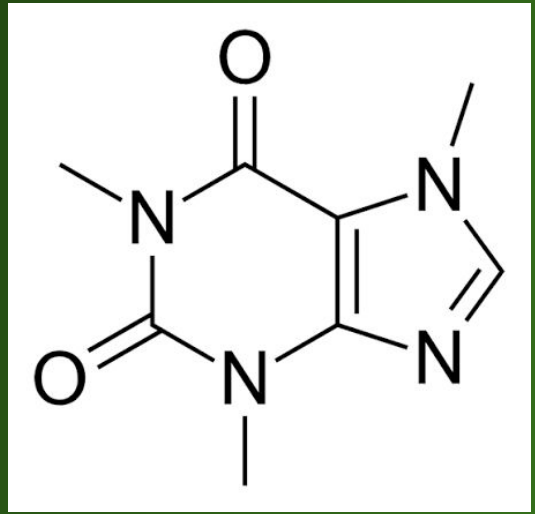
- Structure and charge
- Antiparallel orientation
- 5’ to 3’
- Notations (e.g. 7MGpppApGpUp or dApdTpdC)
- DNA-double stranded, RNA-single stranded

The Molecular Nomenclature

- Nucleosides vs. Nucleotides
- Adenine, Cytosine, Guanine, Thymine, and Uracil vs Adenosine, Cytidine, Guanosine, Thymidine, and Uridine
- AMP vs dAMP
- Purines vs Pyrimidines
- A quick look at the structure...

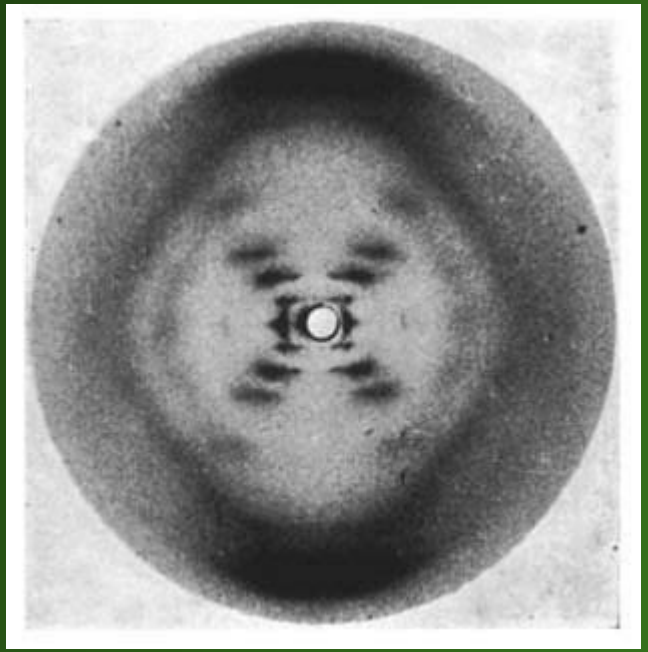
Components of Nucleic Acids

	DNA only	DNA & RNA			RNA only
Nitrogen Bases	 Thymine	 Adenine	 Guanine	 Cytosine	 Uracil
Sugars & Phosphate	 2-Deoxyribose	 Phosphate			 Ribose



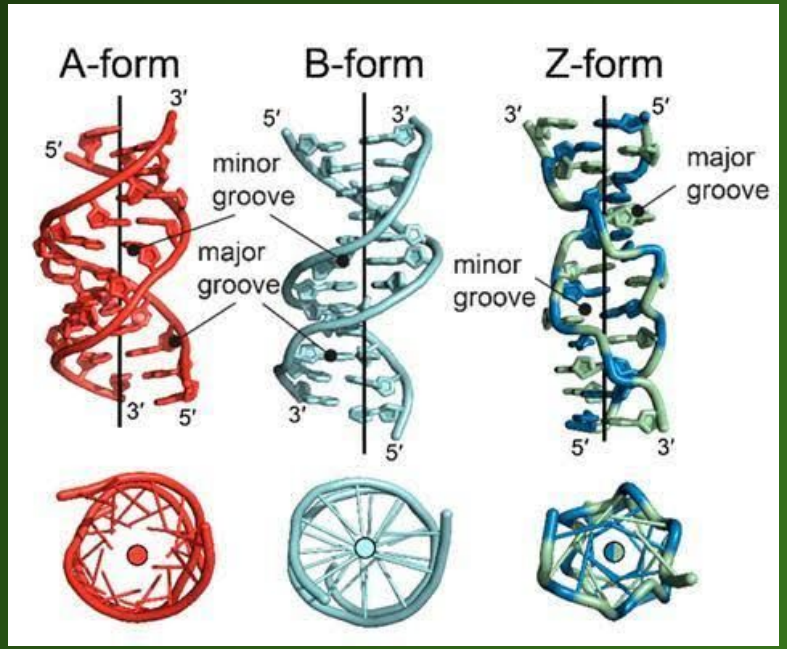
Base structure

- Hetero-cyclic aromatics
- Stable
- Watson-Crick Base pairings
- Rosalind Franklin established the structure
- Total number of T=A, C=G



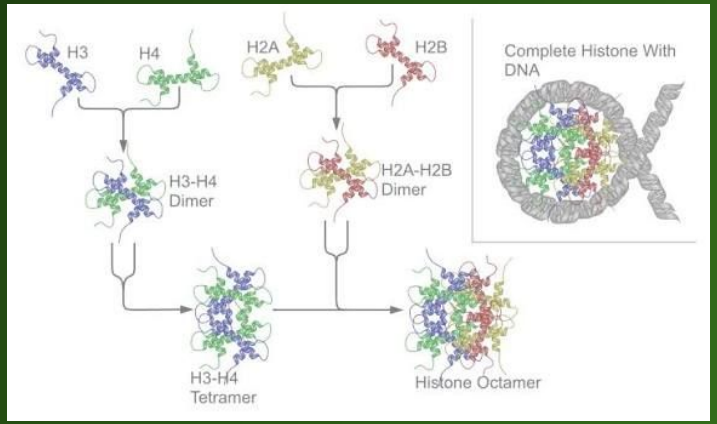
DNA Conformations

- B-Form (10b per turn, 3.4 nm, right handed)
- Z-Form (12b per turn, 4.6 nm, left handed)
- Major and Minor groove
- Denaturation and Annealing



Chromosome and Chromatin Organization

- 46 chromosomes in somatic cells
- Compacting DNA requires Histones and Supercoiling
- Nucleosomes are made up of 2 copies of H2A, H2B, H3, H4, all forming a spindle for DNA to wrap around
- H1 seals the DNA off at the end and holds nucleosomes together

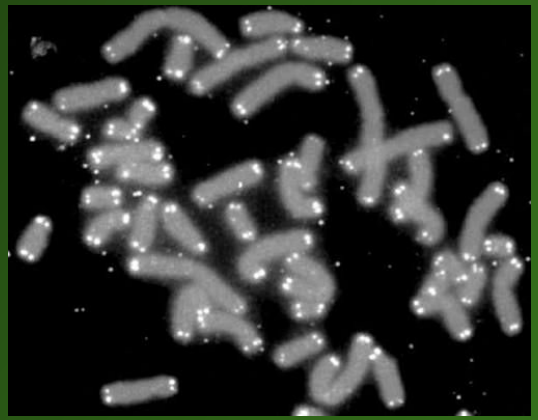


Hetero vs Eu Chromatin

- Expression is dependent on how available DNA is to RNA polymerase
- More tightly wound DNA is less available and is also less permeable to light
- These sections are rarer, and are different, hence hetero-chromatin
- Euchromatin is much more common and is less tightly wound, appearing lighter under a microscope
- Euchromatin is active DNA

Telomeres and Centromeres

- Telomeres are sequences at the ends of a chromosome that act as a fuse and ensure that the ends stay together via a high GC content
- Parts of a telomere are lost in every round of replication, and can be replaced by telomerase
- Centromeres are also high GC regions and are composed of heterochromatin



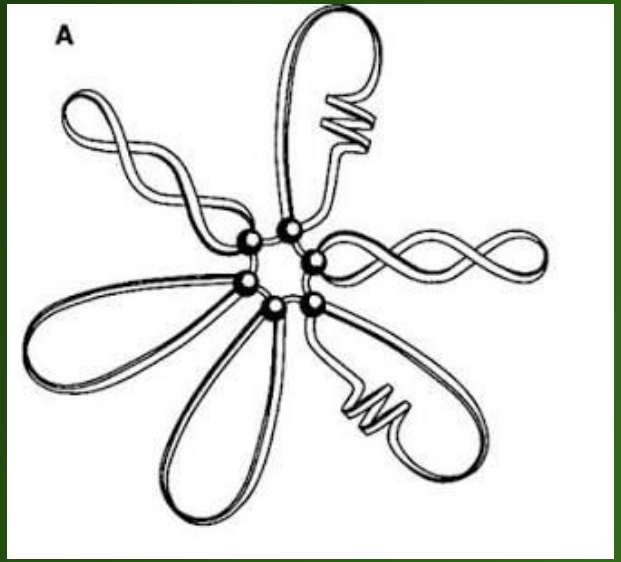
DNA replication

- Eukaryotes have multiple origins of replication, creating many forks
- Linear chromosomes are held together at the centromere following replication
- **Both replication forms are semiconservative**

- Prokaryotes have single origin in a circular chromosome, which creates two forks
- Each synthesized chromosome is separate after replication

Replication initiation Machinery

- Helicase
- Single-Strand DNA binding proteins
- Topoisomerases (Supercoiling)



Replication Machinery cont

- DNA polymerase
  - Lagging and Leading strand synthesis
    - Okazaki Fragments
    - Primer dependent
    - Alpha through Epsilon in Eukaryotes, 1-5 in Prokaryotes
      - Gamma works in mitochondria
      - Delta fills in gaps after primers are removed
      - Alpha, Delta and Epsilon work in DNA replication
      - Beta and Epsilon are used in DNA repair
      - Delta and Epsilon use PCNA protein to bind closely to DNA
- Primase
  - Only a few for leading strand
  - Several for lagging strand
- RNaseH or DNA polymerase 1 5'-3' exonuclease
- Ligase

DNA Damage and repair

- Oncogenes and Tumor suppressor genes
- Proofreading via 5’-3’ exonuclease
- Mismatch repair via MSH2 and MLH1 in Eukaryotes, MutS and MutL in prokaryotes during G_2
- Nucleotide excision repair vs base excision repair
    - Excision endonuclease + DNA polymerase
    - Glycolase + AP endonuclease + DNA polymerase
    - Thymine dimers
    - Adenine deamination

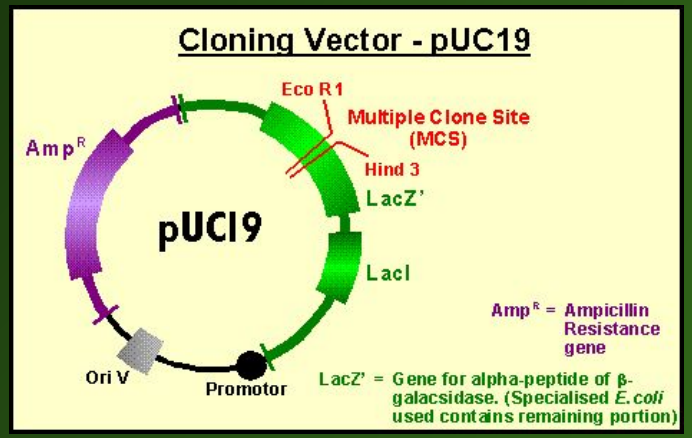
Polymerase Chain Reaction

- Denaturation
- Annealing
- Extension
- Relies on Polymerase and probe DNA to amplify existing DNA



Cloning and DNA vectors

- Restriction enzymes
- Bacterial cloning
- Induction or plasmid purification
- Plasmids contain antibiotic resistance genes to verify transformation

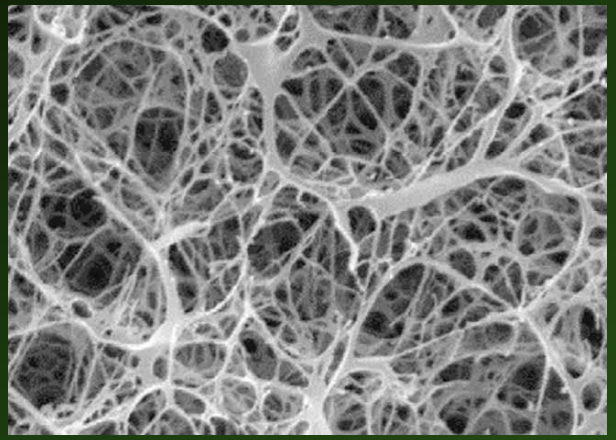


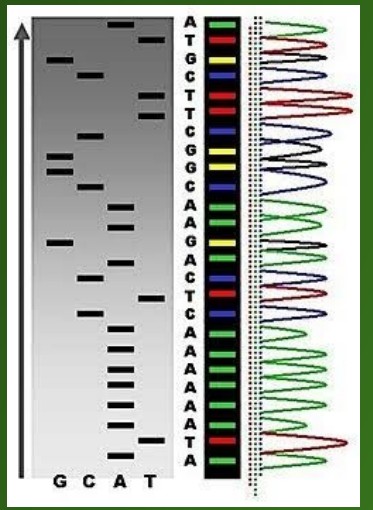
DNA and cDNA libraries

- Genomic
  - Contains introns and exons, fragmented into various vectors
  - May not have full genes represented
- Complementary
  - Contains reverse transcription of expressed DNA (i.e. mRNA)
  - Has full expressed genes for proteins but lacks introns

Gel Electrophoresis, Southern Blotting

- Agarose matrix with a charge gradient to separate samples by size
- Southern blotting uses indicator complementary DNA to test for presence of certain sequence
- Sanger Sequencing uses PCR fragments, created via ddNTPs to sequence DNA, ordering the fragments by weight





Gene Therapy and Transgenic Mice

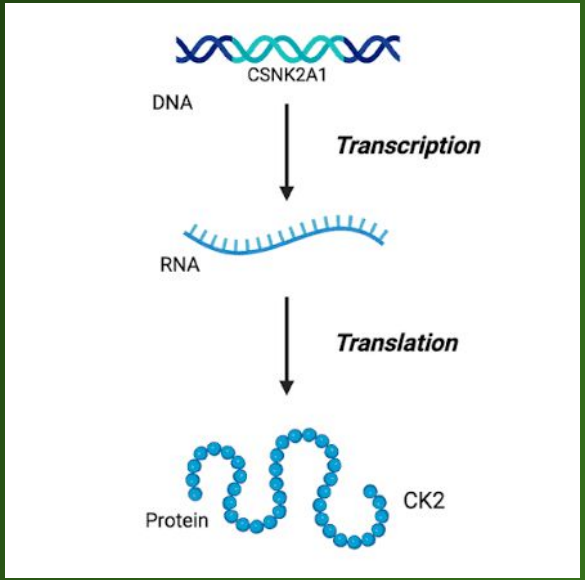
- By using retrovirus vectors to incorporate DNA into the genome of a target, recessive diseases can be treated
- Targeting germ cells of mice breeds can be a useful animal model for various expression patterns, including knockouts, knockdowns, or high expression models
- Ethics...

RNA and Gene Expression

The Central Dogma

DNA -transcription-RNA-translation-Protein

RNA is synthesized in the same way as DNA, via RNA polymerase (5’-3’)



Types of RNA

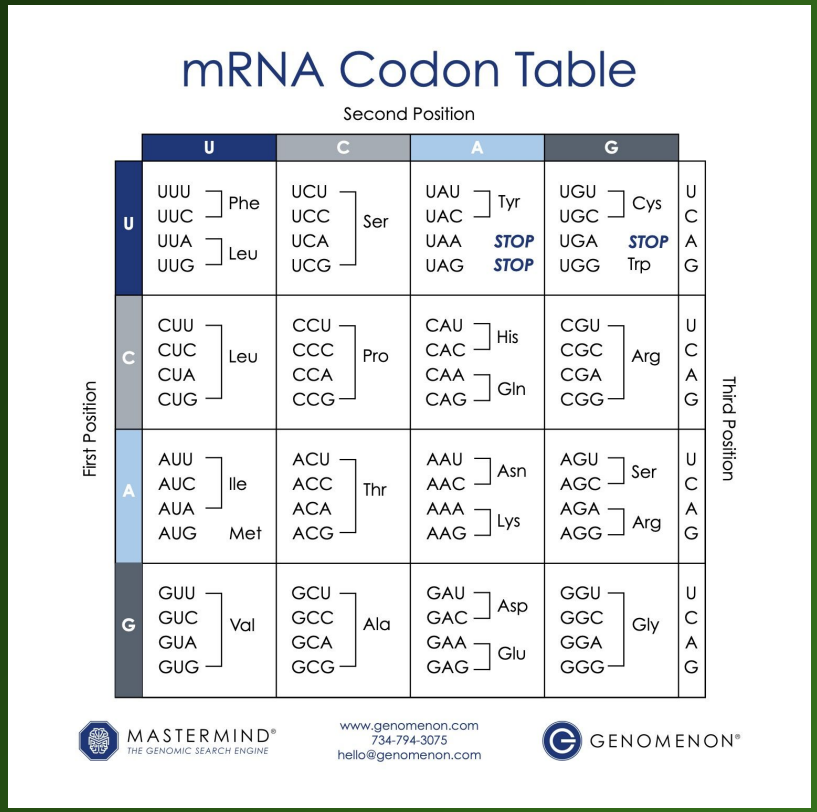
- mRNA-Messenger RNA
  - Synthesized from the genome, and travels from the nucleus to the cytosol
  - Eukaryotes have monocystronic mRNA, prokaryotes have polycistronic mRNA
- tRNA-Transfer RNA
  - Carries amino acids to the ribosome and pairs with mRNA in polypeptide synthesis (Charged/Activated form carries an amino acid) *Still pairs antiparallel to mRNA
- rRNA-Ribosomal RNA
  - Synthesized in the nucleolus, some rRNA molecules make up parts of the ribosome and act as ribozymes
- And more… but don’t worry about that

The genetic language

Note* Start codon (AUG) codes for methionine. Most proteins have a methionine at the N-Terminus for this reason

Degeneracy and mutations

- Nonsense
- Missense
- Silent
- Frameshift



Transcription

- Helicases and topoisomerases assist RNA polymerase in reading the antisense or **template** strand (3'-5') to synthesize mRNA in 5'-3' orientation
- The pre-mRNA looks identical to the **coding** strand of DNA, just with Uracil instead of Thymine
- RNA polymerase exists as I, II, or III. RNA polymerase II synthesizes pre-mRNA
- The first base transcribed is designated as +1, the base before as -1
    - This allows for upstream regions to be described in their proximity to a gene

Post-Transcriptional Processing

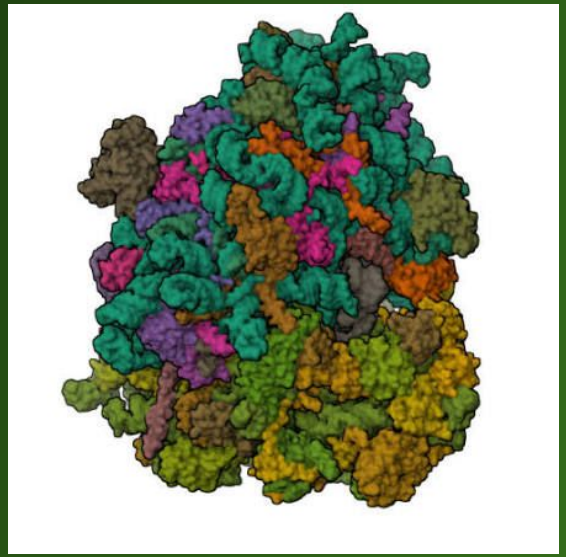
- Splicing
  - Removes noncoding sequences (introns), and ligates coding sequences (exons) together
  - snRNA and SNRPs ‘snip’ introns out, forming a lariat structure that then excludes the intron
- m7Gppp cap
  - At the 5’ end, a cap is added to protect the mRNA from 5’ exonuclease activity
- Poly-A tail
  - A string of Adenines at the end of the mRNA transcript, acting like a fuse for the lifespan of the mRNA molecule

Mature mRNA components

- A m7Gppp cap
- 5’ and 3’ UTRs due to requiring start and end codons
- Exons*
- A poly-A tail

Components of Translation

- mRNA exits the nucleus through nuclear pores
- Ribosomes are made up of proteins and rRNA, and consist of 2 subunits:
  - 40S and 60S to form an 80S ribosome in Eukaryotes and 50S and 30S to form a 70S ribosome in Prokaryotes



Translation

- Initiation
  - Small ribosome subunit binds first
    - In Eukaryotes, to the cap
    - In prokaryotes, to a sequence present in every mRNA synthesized (Shine-Dalgarno sequence)
  - Met/fMet charged tRNA binds to start codon in mRNA
  - Large Subunit binds and translation begins
- Elongation
- Termination
- All steps require specific protein factors and GTP

Translation

- Initiation
- Elongation
  - Ribosomes have 3 catalytic sites used for translation
  - A Site
    - Aminoacyl binding site, binds the next tRNA that pairs to the next codon
  - P Site
    - Peptide bond formation occurs. The growing chain is attached to the amino acid on the A site tRNA as it shifts to the P site via peptidyl transferase + GTP
  - E Site
    - Now uncharged, the P site tRNA binds here briefly before dissociating from the ribosome to be recharged
  - Elongation factors recruit appropriate tRNAs with GTP
- Termination
- All steps require specific protein factors and GTP

Translation

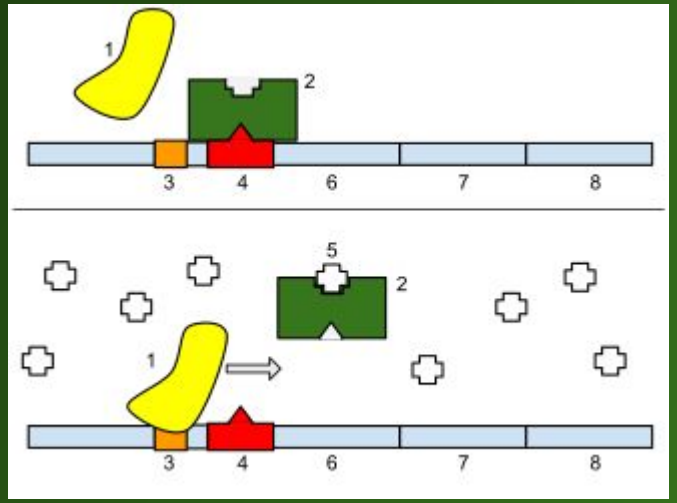
- Initiation
- Elongation
- Termination
  - Signal sequences on the peptide chain can direct the protein to the ER for preparation for secretion or to other subcellular compartments
  - When a stop codon is read, peptidyl transferase hydrolyzes the peptide chain from the last tRNA, freeing the peptide from the ribosome
  - The ribosome than dissociates
- All steps require specific protein factors and GTP

Post Translational Modifications

- After the primary sequence is formed, some proteins require careful management to fold the tertiary structure—chaperone proteins assist in this
- Some proteins require cleavage to become active (insulin, APP)
- Other proteins need cofactors or covalently added groups attached, such as:
  - Phosphoryl
  - Carboxyl
  - Oligosaccharides
  - Lipid groups

Control of Gene Expression

- Prokaryotes
  - Operon structure
    - Promoter
    - Operator
    - Regulator Gene - Repressor protein
    - Gene of interest
    - Small molecular signal



Prokaryotic Gene Expression continued

- Types of control
  - Negative or positive
    - Presence of protein decreases or increases transcription, respectively (Focus on the outcome)
  - Either inducible or repressible
    - Inducible genes are off unless signal is present
    - Repressible genes are on unless signal is present

Any combination of the two are possible

- Negative/positive inducible/repressible operon

Eukaryotic Gene regulation

- Transcription factors
  - DNA binding domain identifies binding motifs in DNA to localize to
  - Activation domain assists in recruiting transcription machinery such as histone acetylases or RNA polymerase
- Distant enhancer elements are activated by molecular signals, which can increase transcription (CREB-cAMP activating transcription)
  - They may be up to 1000 bp away from a gene, which distinguishes them from promoters, which must be around 25 bp away
- Gene Duplication
- Chromatin remodeling via acetylases, deacetylases, methylases, and methyl tranferases